

PROBLEMS ON MECHANICAL PROPERTIES

1. A cylindrical aluminum sample has a diameter of 12.8 mm and an initial base length of 12.5 mm and it is subjected to uniaxial tension. Use the data from the load/elongation table to answer the following questions:

- Represent the engineering stress data versus the engineering strain.
- Calculate Young's modulus. (**Solution: $E \approx 200 \text{ GPa}$**)
- Calculate the yield strength, assuming that the residual strain at the yield point is 0.002. (**Solution: $\sigma_y = 740 \text{ MPa}$**)
- Work out the tensile strength of this alloy. (**Solution: $\sigma_{max} = 1250 \text{ MPa}$**)
- Determine the approximate value of ductility expressed as a deformation percentage. (**Solution: 11.3%**)
- Determine the approximate value of the modulus of resilience. (**Solution: $U_r = 1.37 \text{ MPa}$**)
- Calculate the load (N) and the elongation (mm) of the sample at the yield point. (**Solution: $F = 9.5 \times 10^4 \text{ N}$, $\Delta l = 0.071 \text{ mm}$**).
- Calculate an approximate working stress for this material. (**Solution: $\sigma_w = 370 \text{ MPa}$**)
- With the values obtained, is it possible to calculate the load applied to the sample if the elongation produced is 0.1 mm? If it is possible, work it out and if it is not possible, explain your answer and provide an alternative method. (**Solution: It can only be obtained graphically; $F = 10.3 \times 10^4 \text{ N}$**).

F (N/100)	127	254	381	508	762	891	927	1025	1078	1194	1283	1497	1590	1604	1595	1515	1247	Fracture
ΔL (mm)	0.006	0.012	0.019	0.025	0.038	0.051	0.063	0.095	0.127	0.190	0.254	0.508	0.762	0.889	1.016	1.270	1.460	

2. A cylindrical sample made from a metallic material (assume a Poisson ratio of $\nu = 0.25$) of 15mm in diameter and a calibrated length of 50mm was subjected to a tensile test, and the results are shown in the table below:

Force (kN)	70	120	150	160	170	200	220	233	233	220
Δl (mm)	0.25	0.4	0.5	0.6	0.75	1.75	3.00	5.00	6.50	8.00

The diameter of the sample after fracture was 12.45 mm. Calculate:

- The tensile strength. (**Solution: $\sigma_{UTS} = 1319 \text{ MPa}$**)
- The Young's Modulus (or elastic modulus). (**Solution: $E = 85 \text{ GPa}$**)
- The yield stress. (**Solution: $\sigma_y = 920 \text{ MPa}$**)
- The modulus of resilience. (**Solution: $U_r = 4.98 \text{ MPa}$**)
- The real tension for an engineering deformation of 0.02. (**Solution: $\sigma_r = 1030 \text{ MPa}$**)
- The percentage of deformation to fracture and percentage of area reduction. (**Solution: 15% , 7.3%**)

3. An uniaxial tensile test was performed on a cylindrical sample of 13 mm in diameter getting the results of load and elongation shown in the table below:

- Calculate the engineering stress and strain values and draw the stress-strain curve.
- Calculate the value of the elastic modulus (Young's Modulus) and determine the length of a bar of 13 cm on which a tension of 200 MPa is applied. (**Solution: $E = 75.4 \text{ GPa}$, $l = 13.04 \text{ cm}$**)
- What is the recommended cross-section for a cable that has to support an elevator with a maximum weight of 450 kg? (**Solution: $A_0 \geq 16.9 \text{ mm}^2$**)
- Calculate the real stress and real strain values at maximum load and at the fracture point, assuming that the diameter of the sample is 12.922 mm and 10.348 mm respectively. Do the calculations without using the engineering stress and strain values. (**Solution: Load max.: $\sigma_{UTS} = 300 \text{ MPa}$, $\epsilon_{UTS} = 0.012$; Fracture point: $\sigma_{fracture} = 420 \text{ MPa}$, $\epsilon_{fracture} = 0.456$**)
- Calculate the modulus of resilience. (**Solution: $U_r = 0.457 \text{ MPa}$**)
- Calculate the elastic strain at the moment before fracture. (**Solution: $\epsilon_y = 3.4 \times 10^{-3}$**)

Load (kg)	Calibrated length (mm)	Engineering stress (MPa)	Engineering strain
0	5.148		
500	5.151		
1500	5.156		
2500	5.161		
3500	5.166		
3750	5.225		
3950	5.354		
4000 (maximum load)	5.457		
4000	5.560		
3600 (point of fracture)	5.676		

4. A tensile test was performed on a brass sample with a rectangular section and dimensions of 3.2 mm x 19.1 mm. Using the data in the table below, answer the following questions:

Load (N)	0	1380	2780	5630	7430	8140	9870	12850	14100	14300	13830	12500
Length (mm)	63.50	63.53	63.58	63.63	63.70	63.75	64.14	65.41	66.68	67.95	69.22	70.49
σ_n (MPa)												
ϵ_n												

- Represent the data in terms of engineering stress *versus* strain.
- Calculate the modulus of elasticity (Young's modulus). (**Solution: $E \approx 44 \text{ GPa}$**)
- Determine the yield strength. (**Solution: $\sigma_y = 150 \text{ MPa}$**)
- Determine the tensile strength. (**Solution: $UTS = 234 \text{ MPa}$**)
- Calculate the modulus of resilience. (**Solution: $U_r = 0.28 \text{ MPa}$**)
- Calculate the working stress of this material. (**Solution: $\sigma_w = 75 \text{ MPa}$**)

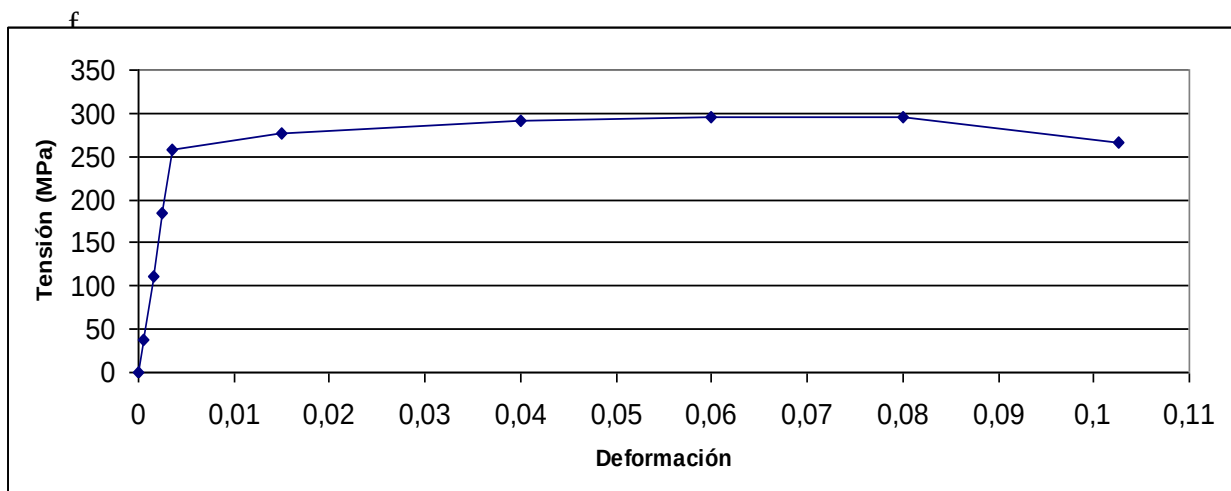
5. A steel wire, initially 1 meter in length, is stretched by a force of 1400 N at a temperature of 20 °C. After the force is applied, the wire is held at a constant length and then heated to 40 °C. Determine the required diameter of the wire to ensure that the thermal expansion stress compensates for the applied stress. **(Solution: 5.86 mm)**

Data: $E = 207 \text{ GPa}$; $\alpha_l = 12.5 \cdot 10^{-6} \text{ }^\circ\text{C}^{-1}$;

6. A sample made of a Ni alloy with an elastic modulus of 207 GPa and an initial diameter of 10.2 mm experiences an elastic deformation only when loads inferior to 8900 N are applied. Calculate the maximum length of the sample before the deformation if the maximum elongation allowed is 0.25 mm. **(Solution: $l_0 = 475 \text{ mm}$)**

7. In a construction site, a change in the diameter of one of the crane cables was detected. When the weight that the cable was withstanding was 48950 N, the change was $5.3 \cdot 10^{-3} \text{ mm}$, and when the load was removed, the cable recovered its initial diameter of 15 mm. Considering that for this kind of steel the Poisson's ratio $\nu = 0.33$ and according to the manufacturer data the initial length of the cable is 3 m.

- Calculate the length of the cable when it's holding the load mentioned above. **(Solution: 3003.2 mm)**
- If it has been detected that plastic deformation of the cable begins when it is holding a weight of 10000 kg, can it be considered that the cable is working under safe conditions? Assume a safety factor of 2. **(Solution: yes)**
- What is the value of the elastic modulus for this material? **(Solution: $E = 261.3 \text{ GPa}$)**
- In order to prevent service problems the provider replaces the cable for a different one, which has a stress-strain curve represented below and an elastic modulus of $E = 75 \text{ GPa}$. Can this material replace the previous one? Explain why. **(Solution: no)**
- Calculate the final length of this new cable in case it is working under the same conditions (48950 N). **(Solution: 3045 mm)**



8. For a particular kind of brass, plastic deformation begins at 345 MPa, and its elastic modulus is 103 GPa.
- What is the value of the yield stress? (**Solution: $\sigma_y = 345 \text{ MPa}$**)
 - What is the maximum stress that can be applied to a sample of this brass with a cross-section 130 mm^2 without plastic deformation taking place? And the maximum load? (**Solution: $\sigma_{max} = 345 \text{ MPa}$, $F_{max} = 44850 \text{ N}$**)
 - If the initial length of the sample is 76 mm, what is the maximum length that it can reach without plastic deformation? (**Solution: $l = 76.25 \text{ mm}$**).
9. A load of 48900 N produces elastic deformation in a cylindrical sample of 15.2 mm in diameter and a length of 254 mm. Considering that for this kind of steel $E = 207 \text{ GPa}$ and $\nu = 0.27$, determine the elongation in the direction of the applied force and the change in diameter of the sample. (**Solution: $\Delta l = 0.33 \text{ mm}$, $\Delta \phi = -0.0053 \text{ mm}$**).
10. Plastic deformation in a copper wire of 200 cm in length and a diameter of 1 mm begins when it is loaded with 15.3 kg. If the elastic modulus for copper is 110 GPa.
- Calculate its yield strength. (**Solution: $\sigma_y = 190.9 \text{ MPa}$**)
 - Considering that when it is loaded with 29.6 kg the strain is 0.011, calculate the final length of the wire once it has been unloaded. (**Solution: $l = 201.85 \text{ cm}$**)
 - Calculate the resilience modulus. (**Solution: $U_r = 0.165 \text{ MPa}$**)
11. A cylindrical steel sample with a diameter of 15 mm and a length of 250 mm experiences elastic deformation when a load of 50 kN is applied. Considering that for this steel $E = 207 \text{ GPa}$ and $\nu = 0.27$, determine the elongation in the direction of the force applied and the change in diameter of the sample. (**Solution: $\Delta l = 0.342 \text{ mm}$, $\Delta \phi = -0.0055 \text{ mm}$**)

12. A steel cable has fracture strength of 300 MPa. Its ductility is 77% measured as a reduction in area. Calculate the real fracture strength. Considering that strain is uniform until fracture, what is the real strain? **(Solution: $\epsilon_{real} = 1.47$)**

13. Calculate the Vickers and Brinell hardness numbers for a metallic material on which several hardness tests were performed, obtaining the following results:

- a. When the diamond Vickers hardness test with a load of 30 kg was performed, the average length of the diagonal was: first impression 0.527 mm, second impression 0.481 mm, third impression 0.497 mm.
- b. When using the Brinell test with a ball of 10 mm and a load of 3000 kg, the diameter of the indentation was 4.01 mm.

(Solution: 220 HV, 227 HB)

14. A Brinell indenter of 10 mm in diameter produced an indentation of 2.5 mm in diameter in a steel when a load of 1000 kg was applied. Calculate the Brinell hardness for that material. **(Solution: 200 HB)**

15. What will be the diameter of the indentation that produces a Brinell hardness of 300 when a load of 500 kg is applied? Diameter of the ball: 10 mm. **(Solution: $d = 1.453 \text{ mm}$)**